

Partha Sarathi Banerjee

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Profile

Quantitative risk professional with a PhD in Theoretical Physics and experience in financial services risk management at Ernst & Young, with prior experience in model risk management at Morgan Stanley. Experienced in market risk RWA projections under ICAAP using VaR-based and FRTB Standardised Approach methodologies, interest rate curve construction and interpolation (OIS) using SOFR, €STR and SONIA, prudent valuation under ICAAP stress scenarios, and validation of third-party models in non-financial risk, with a strong focus on quantitative modeling applied to financial risk.

Experience

Senior Consultant, FSRM, Ernst & Young LLP March 2026 – Present
Mumbai, Maharashtra, India

- Developed Python-based interest rate curve construction routines using bootstrapping and interpolation in multi-curve frameworks for OIS discounting (SOFR, €STR, SONIA).
- Implemented piecewise linear (zero rates, discount factors) and log-linear interpolation methods, evaluating their impact on curve construction and valuation.

Associate (intern), Model Risk Management, Morgan Stanley Jan 2025 – July 2025
Mumbai, Maharashtra, India

- Conducted independent revalidation of Tier 1 market risk models for Market Risk RWA under ICAAP and assessed limitation closure for Prudent Valuation under ICAAP stress, ensuring compliance with internal model governance and regulatory standards within a Model Risk Management (MRM) framework.
- Performed model validation using backtesting, benchmarking, stress testing, and sensitivity analysis; assessed assumptions, limitations, and methodologies across VaR-based frameworks (VaR, Stressed VaR, RNIV) and FRTB Standardised Approach (SA).
- Evaluated model risk in AI/ML identity verification models, analysing data quality, assumptions, and performance metrics; documented findings and coordinated remediation with model developers.

Skills

Programming Languages: Python 3+ (NumPy, Pandas, SciPy, Matplotlib, Seaborn), FORTRAN, C, R, Julia.

Quantitative Finance: Value at Risk (VaR), Stressed VaR, Risk Not in VaR (RNIV), RWA calculation with FRTB regulations, Prudent Valuation, Derivative Pricing.

Numerical Methods: Monte Carlo, Finite Difference Method, Linear regression, Ridge regression, Lasso regression, Elastic Net, PCA regression.

Computational Experience: 8+ years of experience implementing mathematical models in code; performed large-scale simulations on a high-performance computing (HPC) environment.

Tools: LaTeX typesetting, Mathematica, MATLAB, MS Excel.

Operating Systems: Linux, Windows.

Other Technical Skills: Stochastic Calculus, Statistical Analysis and Research Methodologies.

Languages: English (Proficient), Bengali (Native), Hindi (Proficient).

Education

Ph.D. Physics 2019 – 2026
Indian Institute of Technology Delhi, India

Specialization: Quantum Condensed Matter Physics

Thesis title: Scattering of massless Dirac fermions in modulated graphene junctions and its effect on electron transport

M.Sc. Physics

2017 – 2019

Indian Institute of Technology Delhi, India

Thesis title: Parametric resonance in presence of noise and dissipation memory kernel.

CGPA Obtained: 8.512

B.Sc. Hons. in Physics

2014 – 2017

Bangabasi College, University of Calcutta, India

Academic Visit

Visiting Researcher (SPARC Project)

Nov 2025 – Jan 2026

As part SPARC project under the supervision of Prof. Efrat Shimshoni at Bar-Ilan University, Ramat Gan, Israel.

Awards and Achievements

- 2019: Qualified JEST and GATE for PhD.
- 2018: Qualified Joint CSIR-UGC NET(LS).
- 2018: Qualified JEST for PhD (Rank 125).
- 2017: State Topper (West Bengal) of National Graduate Physics Examination of IAPT.
- 2017: College Topper in BSc(Hons.) Examination at Bangabasi College, Kolkata, India.
- 2017: Qualified JEST for Integrated PhD (Rank 210) and IIT-JAM for M.Sc (Rank 109).

Project / Internship

MSc Project

July 2018 - July 2019

“Studying Parametric Resonance in the presence of Noise and Dissipation Memory Kernel” under the supervision of Prof. Rahul Marathe at Indian Institute of Technology, Delhi, India.

Summer Research Internship

May 2018 - July 2018

“Aubry Transition in Colloidal Monolayers” under the supervision of Dr. A. V. Anil Kumar at the National Institute of Science Education and Research (NISER), Bhubaneswar, India

Research Publications

1. P. S. Banerjee, R. Marathe, and S. Ghosh, *Intermediate chiral edge states in quantum hall josephson junctions* (submitted, 2025), arXiv:2510.11432.
2. P. S. Banerjee, R. Marathe, and S. Ghosh, “Electronic analogue of fourier optics with massless dirac fermions scattered by quantum dot lattice,” *J. Opt.* **26**, 095602 (2024).
3. P. S. Banerjee, R. Marathe, and S. Ghosh, “Magnetically modulated superconductor-graphene-superconductor (SGS) josephson junctions and their tunability,” *Phys. Scr.* **100**, 015965 (2024).

Conferences Attended

- Quantum Materials in the Quantum Information Era from 25 - 29 September 2023 at the Max Planck Institute for the Physics of Complex Systems (MPIPKS), Dresden, Germany.
- Emergent phenomena in van derWaals heterostructures 2023 from 9-12 Jan 2023 at TIFR, Mumbai, India.

Talks and Presentations

- “Intermediate chiral edge states in quantum Hall Josephson junctions,” Condensed Matter Seminar, Racah Institute of Physics, Hebrew University of Jerusalem, Israel, 15 Jan 2026